

George David Tsitlauri

AI & SYSTEMS RESEARCH ENGINEER

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BSc in Informatics & Telecommunications, University of Thessaly, Greece (2021-2026)

MSc in Artificial Intelligence & Applications, University of Thessaly, Greece (2026-2027)

Specialization: Data Management and Artificial Intelligence

Research Laboratory: Computer Architecture, Compilers and System Security

RESEARCH PROJECTS

AI, DATA & INTELLIGENT SYSTEMS

KRONOS - Causal domain-invariant learning algorithm that enables AI models to generalize reliably across unseen domains and conditions.

NOUS - Self-improving LLM research framework for iterative refinement, reasoning loops, and resource-constrained experimentation.

VALKYRIE - Real-time video and audio enhancement framework optimized for consumer GPUs.

PATV-X - Training-free explainable deepfake-video detection framework using frequency-domain analysis and forensic signal cues.

SIREN - Adversarial-audio and voice-forensics framework for spoofing analysis, deepfake detection, and audio-security evaluation.

ATHENA - Precision-medicine and biomedical-informatics framework integrating genomics, health records, biomedical knowledge graphs, and causal reasoning.

ASCLEPIUS - Biomedical AI framework for multimodal biosignal analysis, anomaly detection, explainability, and federated-health experimentation.

SOCRATES - Adaptive education research framework that models why learners struggle and dynamically adjusts learning paths using causal tutoring algorithms.

THEMIS - Hybrid search framework combining classical and neural retrieval with causal reranking to improve result relevance across text and multimedia queries.

MarketMindAI - Behavioral-finance research framework linking internet sentiment and NLP signals to cryptocurrency market dynamics.

EVOO - Applied machine-learning framework for olive-oil authenticity and adulteration analysis with spectroscopy data.

CYBERSECURITY, SYSTEMS & INFRASTRUCTURE

OLYMPUS - AI security research platform covering offensive, defensive, forensic, and intelligence modules — with co-evolutionary attacker/defender simulation, reverse engineering, and malware analysis.

CAESAR - Standalone co-evolutionary engine for adversarial strategy search, where attacker and defender populations evolve independently to discover non-obvious attack/defense equilibria.

LYKOMIDES - Cross-domain AI intrusion-detection framework designed to transfer across cyber-physical and IoT environments without retraining on target data.

CUDA-zkML - GPU-accelerated zero-knowledge machine-learning system for verifiable neural-network inference.

ATLAS - Distributed cloud-autonomy framework combining digital twins, orchestration, governance, and intelligent control.

NEXUS - End-to-end systems research platform spanning compiler construction, MIPS code generation, CPU simulation, parallel execution, and hardware validation.

APOLLO - From-scratch Java database-engine research project covering storage, B+-tree indexing, MVCC transactions, and an ML-learned query optimizer.

PYTHAGORAS - Unified computational theory and discrete mathematics research framework covering the full Chomsky hierarchy, Turing machines, NP-completeness, and decidability, with the LOGOS formal verifier.

POLARIS - SDN/cloud orchestration research platform for multi-objective service placement and network-aware optimization.

CHRONOS - Edge-IoT scheduling and offloading research framework using causal structures and multi-objective decision policies.

ARGUS - Game-theoretic distributed microkernel research with Nash-equilibrium scheduling and lock-free IPC for edge-AI workloads.

ARTEMIS - Mobile reliability research framework for proactive fault detection and SLO monitoring that identifies service degradation before users are affected.

HARDWARE, TELECOM & ROBOTICS

ODYSSEY - Hybrid quantum-classical security research platform for post-quantum reasoning and risk-aware intrusion analysis.

Crypto3DStackCPU - Secure CPU-extension research project combining 3D-stacked memory with hardware cryptographic extensions, custom assembler, and runtime integrity validation.

DAEDALUS - Evolutionary robotics and optimization framework for co-designing morphology, controllers, and adaptive behaviors.

HERMES - AI-native software-defined radio framework for learned modulation, channel adaptation, and intelligent signal processing.

GATE-6G - Semantic-communication and split-inference research framework for adaptive 6G-style AI-native transmission.

HELIOS - Multi-language signal-processing framework (Python, Julia, R, Octave) covering arithmetic coding, rate-distortion theory, and Markov-channel modeling.

IRIS - Neural rendering research framework for real-time perception and visualization on consumer GPUs.

ARES - Robotics autonomy research platform for control, planning, navigation, and multi-agent coordination experiments.

KLADENETS - Autonomous UAV-swarm research architecture centered on edge AI, resilience, and explainable decision-making.

POSEIDON - Multi-domain maritime systems research framework combining GIS, CAD, optical networks, and TRIDENT optimization.

HEPHAESTUS - RTL and physical-design research on power-aware INT8 MAC architectures for edge-AI workloads, with measured power reduction via the CYCLOPS algorithm (SkyWater-130nm).

PROJECTS

ENGINEERING

CryptoCPU (BSc Thesis) - Design and implementation of an FPGA-based encrypted instruction-set processor.

Parallel AES CUDA - Parallel implementation of AES encryption and decryption on NVIDIA GPUs.

CCC IR Optimizer - Intermediate-representation optimizer with analysis passes and compiler optimizations for the CCC toolchain.

GPBFT Consensus Protocol - Distributed consensus implementation inspired by Byzantine agreement and signature-based coordination.

Build a Compiler - Compiler project covering frontend parsing, AST construction, semantic analysis, and MIPS-oriented code generation.

AI / DATA IMPLEMENTATIONS

Advanced AI Projects - Collection of advanced machine-learning and deep-learning implementations across vision, generation, and optimization tasks.

Weather App Autoscaler - Kubernetes-based weather application with autoscaling and load-aware deployment behavior.

Big Data Spark Taxi - Distributed Spark pipeline for large-scale taxi-trip analytics.

Video Coding for Machines - Video-coding experiments tailored to machine-learning and machine-vision applications.

Reinforcement Learning Experiments - Reinforcement-learning implementations and experiments across core algorithms and environments.

Tourism Data Mining - Data-mining project for tourism datasets, trend analysis, and pattern discovery.

Egram Student Management System - Student-management platform for academic administration, records, and workflow handling.

WEBSITES

Skyrianos - Public-facing website about the island of Skyros with informational content and visitor-oriented presentation.

OstrichZooPark - Public-facing website for Ostrich Zoo Park with visitor information and general presentation.

Pavlidis Home E-shop - E-commerce website for online product presentation and storefront workflows.

SKILLS

Programming Languages	Python, C, C++, Java, SQL, JavaScript, TypeScript, Assembly, Go, Rust, Haskell, Dart, VHDL, Verilog, C#, Scala, Julia, R, MATLAB, Octave, Kotlin, Swift, Solidity, GLSL, Fortran, PHP, Ruby, Perl, Pascal
AI / Data	PyTorch, TensorFlow, XGBoost, Apache Spark, Apache Hadoop
Systems / Software	FastAPI, Flutter, Streamlit, Docker, Kubernetes, PostgreSQL
Cloud / MLOps	MLflow, Weights & Biases, AWS Lambda, AWS SageMaker, Azure Functions, Azure Blob Storage, Azure ML, GCP BigQuery, Google Cloud Storage, Cloud Functions
Hardware / Parallel Computing	CUDA, OpenMP, MPI, TensorRT, Xilinx Vivado, Vitis HLS, FPGA/RTL Workflows, HPC
Graphics / XR	Neural Rendering, WebXR, Three.js
Web / Frontend	React, Next.js, Node.js, Tailwind CSS, HTML5, CSS3
Dev / Research Tools	Git, CI/CD, CMake, Jupyter, Wireshark, Bash, PowerShell, Linux, Windows, Visual Studio, LaTeX
Languages	Greek (Native), Russian (Native), English (B2, ECCE - University of Michigan)